

CSA0711
SIMATS ENGINEERING
ASSESSMENT TOOL 3

(6)

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I Ch. Abhilash Reddy (Name / Reg No) CL925284622 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ch. Abhilash Reddy

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Problem Identification	20%	1.0	1.2	1.2	1.2	1.0
Debugging	25%	1.0	1.2	1.2	1.2	1.0
Optimization	20%	1.5	1.5	1.5	1.5	1.25
Analysis	20%	1.2	1.2	1.2	1.2	1.0
Code Quality	15%	0.9	0.9	0.9	0.9	0.75

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Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

CO2 - AT3

Debugging and optimisation tasks -

Debugging the IPv4 workstation configuration.

Given configuration:-

IP address : 192.168.10.65

Subnet mask : 255.255.255.192

Prefix length : /26

Default gateway : 192.168.10.1

Intended subnet : 192.168.10.0/26

Q1:- Determine the subnet size

A/26 Subnet uses 26 bits for the network and 6 bits for hosts.

$$2^{32-26} = 2^6 = 64$$

The subnet mask is:

255.255.255.192

Block size

$$256 - 192 = 64$$

The /26 subnets inside 192.168.10.0/24 are:

192.168.10.0 - 192.168.10.63

192.168.10.64 - 192.168.10.127

192.168.10.128 - 192.168.10.191

192.168.10.192 - 192.168.10.255

Identify the subnet of the configured IP

The workstation IP is

192.168.10.65

This address falls in the range:

192.168.10.64 - 192.168.10.127

Therefore, the workstation actually belongs to

192.168.10.64/26

Steps:-

Network information for the actual subnet

For 192.168.10.64/26:

Network address 192.168.10.64

First valid host: 192.168.10.65

Last valid host: 192.168.10.126

Broadcast address: 192.168.10.127

Verify the gateway:-

The configured gateway is:-

192.168.10.1

The gateway belongs to:-

192.168.10.0/26

Its subnets range is

192.168.10.0/26

Its subnet range is:

192.168.10.0 - 192.168.10.63

Workstation subnet:- 192.168.10.64/26

Gateway subnet:- 192.168.10.0/26

Configuration errors:-

The errors are:-

The workstation IP 192.168.10.65

The gateway 192.168.10.1

Because of this subnet mismatch,

Debugging the JPLIG configuration

Given address:

Device A:

2001 : db8 : %FF00 % u2 : 8329 / 64

Device B:

2001 : db8 : %FF00 % u2 : 8329

Visible groups are:

2001
db8
%FF00
u2
8329

Expand address:

2001 : 0 db8 : 0000 : 0000 : 0000 : %FF00 : 0042 : 8329

Expand device B :-

device B is:

2001 : db8 : 0000 : 1 : 1

Visible groups are:

Network prefix :-

2001 : 0 db8 : 0000 : 0000 : 0001 : 0000 : 0000 : 0001

2001 : 0 db8 : 0000 : 0000 : 164

compressed form:

2001 : db8 : 164

Verify the whether both devices are in the same subnet

Device

Network prefix

Device A

2001:db8::1/64

Device B

2001:db8::2/64

Both the devices are belongs to the IPv6 subnet

Optimized IPv6 configuration:-

Device A

2001:db8::1/64

Device B

2001:db8::2/64

Router or gateway

2001:db8::FFFF/64

corrected configuration:

Device

IPv6 address

prefix

Device A

2001:db8::1

/64

Device B

2001:db8::2

/64

Gateway

2001:db8::FFFF

/64

Number of address in a /64 subnet

$$128 - 64 = 64$$

²⁶⁴
18, 446, 744, 073, 709, 551, 616 address.

IPv4 Subnet Allocation and VLSM Optimization

Given Network

192.168.1.0/24

1/24 network contains:

$$2^8 = 256 \text{ total address}$$

$$256 - 2 = 254$$

Address capacity of Each Subnet

Prefix	Subnet mask	Total address	Usable hosts
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14

Possible /26 subnets

Network address

192.168.1.0/26

192.168.1.64/26

192.168.1.128/26

Valid host range

192.168.1.1 - 192.168.1.62

Broadcast

192.168.1.63

192.168.1.65 - 192.168.1.126

192.168.1.127

192.168.1.129 - 192.168.1.190

192.168.1.255

Possible /27 subnets

Network address

192.168.1.0/27

192.168.1.32/27

192.168.1.64/27

Valid host Range

192.168.1.1 - 192.168.1.30

192.168.1.33 - 192.168.1.62

192.168.1.33 - 192.168.1.94

Broadcast

192.168.1.31

192.168.1.63

192.168.1.95

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Possible /28 subnets

- 192.168.1.0
- 192.168.1.16
- 192.168.1.32
- 192.168.1.48
- 192.168.1.64
- 192.168.1.80
- 192.168.1.96
- 192.168.1.112
- 192.168.1.128
- 192.168.1.144
- 192.168.1.160
- 192.168.1.176
- 192.168.1.192
- 192.168.1.208
- 192.168.1.224
- 192.168.1.240

Examples:-

Network address	Valid host range	Broadcast
192.168.1.0/28	192.168.1.1 - 192.168.1.4	192.168.1.5
192.168.1.16/28	192.168.1.17 - 192.168.1.30	192.168.1.31

calculating unused address

$$\text{unused addresses} = \text{usable capability} - \text{required}$$

Department	Required hosts	Subnet	Usable host
Department A	50	192.168.1.0/26	62
Department B	25	192.168.1.64/27	30
Department C	12	192.168.1.128/28	14
Department D	10	192.168.1.192/28	14
Department E	5	192.168.1.208/29	6

Final answer:-

Use /26 for departments requiring up to 62 hosts, /27 for departments requiring up to 30 hosts, /28 for the departments requiring up to 14 hosts and /29 for departments.

IPv6 configuration debugging

Expanded address:-

Device A:

2001:0db8:0000:0000:0000:ff00:0002:8329/64

Device B:

2001:0db8:0000:0000:0000:0000:0001:64

Network prefixes:

2001:db8::/64

Addressing mismatch

There is no addressing or /64 prefix mismatch in the information provided. The failure is likely related to:

* Firewall blocking ICMPv6

* Incorrect VLAN configuration

* IPv6 disabled on one device

* Duplicate address

* Incorrect interface assignment

Recommend address:

Device A: 2001:db8::1/64

Device B: 2001:db8::2/64

Gateway: 2001:db8::ff/64

Number of address

$$2^{64} = 18,446,744,073,709,551,616$$

corrected /64 :-

18,446,744,073,709,551,616 IPv6 address

Routing Table debugging

Destination network

Next hop

192.168.1.0/24

10.0.0.2

192.168.2.0/24

10.0.0.3

Default route

10.0.0.0

Destination IP :-

192.168.2.100

Apply longest prefix matching

The router compares the destination IP with each routing entry

First route :-

192.168.1.0/24

Range :-

192.168.1.0 - 192.168.1.255

Second route

192.168.2.0/24

the selected route is

192.168.2.0/24

- Determine the next hop

10.0.0.3

Therefore :-

Packets to 192.168.2.100 should be forwarded
to 10.0.0.3

Verify the destination address

from 192.168.2.0/24 with in host range

192.168.2.100 is a valid IP in 192.168.2.0/24

Therefore:-

the default route through 10.0.0.1 will be
selected.

Final answer:-

For destination 192.168.2.100, the longest prefix
match is 192.168.2.0/24, and the packet is forwarded
to 10.0.0.3